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Journal of Ethnopharmacology 80 (2002) 181–186

Journal of ETHNO-PHARMACOLOGY

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Anti-inflammatory, analgesic activity and acute toxicity of *Glaucium grandiflorum* extract

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Received 28 August 2001; received in revised form 2 January 2002; accepted 17 January 2002

Abstract

The species of *Glaucium* have been used in Iranian herbal medicine as laxative, hypnotic, antidiabetic agents and also in the treatment of dermatitis. The anti-inflammatory and analgesic effects of the aerial parts of *Glaucium grandiflorum* Boiss & Huet (Papaveraceae), a native plant of Iran, were studied using carrageenan induced edema, formalin and hot plate tests. The *G. grandiflorum* extract at the dose of 200 mg/kg had more edema inhibition than indomethacin at the doses of 10 ($P < 0.01$) and 8 mg/kg ($P < 0.001$) in the carrageenan test. The ED_{50} (i.p.) in the edema induced by carrageenan was 13.59 mg/kg. In formalin test, the extract (60–90 mg/kg, i.p.) caused graded inhibition of both phases of formalin-induced pain. In hot plate test, the i.p. administration of the extract at the doses of 60, 70, 80 and 90 mg/kg significantly raised the pain threshold at a observation time of 45 min in comparison with control ($P < 0.001$). The extract, at the antinociceptive doses, did not affect motor coordination of animals when assessed in the rotarod model. The 72 h acute LD_{50} value of this extract after i.p. administration in mice was 797.94 mg/kg. © 2002 Published by Elsevier Science Ireland Ltd.

Keywords: *Glaucium grandiflorum*; Anti-inflammatory activity; Analgesic activity; Rotarod test; Acute toxicity

1. Introduction

Glaucium grandiflorum Boiss & Huet (Papaveraceae) is a perennial herb indigenous to various regions of the Mideast extending from the eastern Mediterranean to Iran (El-Afifi et al., 1986). The plant blooms from May until the end of July. The height of the plant is 30–50 cm, petals usually dark orange to crimson, with a darker spot at the base (Davis, 1967; Reschinger, 1966). The species of *Glaucium* have been used in Iranian herbal medicine as laxative, hypnotic and antidiabetic agents. The leaves of the species of *Glaucium* are also used in the treatment of dermatitis (Zargari, 1990). Bibliographical survey showed that a few phytochemical investigations have so far been carried out on this species. Isolation of some alkaloids [(−)-norchelidonine, dihydrochelerythrine, (−)-8-acetonyldihydrochelerythrine, protopine, allocryptopine, (±)-tetrahydrojarrorrhizine(corypalmine), (±)-tetrahydropalmatine, glaucine, corydine, isocorydine, cryptopine, trans-canadine methochloride, berberine] from this species have been reported (El-Afifi et al., 1986; Gozler 1982; Phillipson et al., 1981).

Glaucine shows analgesic and anti-inflammatory activity; tetrahydropalmatine has analgesic activity (Ayres, 1994).

In continuation of studies of Iranian wild species of the Papaveraceae family (Shafiee et al., 1998; Shafiee and Morteza-Semnani, 1998, 1999), *G. grandiflorum* attracted much interest, because, there was no scientific study about its pharmacological effects. Therefore, we decided to investigate anti-inflammatory, antinociceptive activity and acute toxicity of *G. grandiflorum* extract for the first time.

The present paper describes some of the pharmacological activities of this plant using carrageenan-induced rat paw edema, formalin, hot plate and rotarod tests.

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0378-8741/02/$ - see front matter © 2002 Published by Elsevier Science Ireland Ltd.
PII: S0378-8741(02)00027-2

2. Materials and methods

2.1. Plant material

G. grandiflorum Boiss & Huet was collected from the area of Sorkhe-hesar (the suburb of Tehran), in the north of Iran, in spring 1999 and identified by Dr Gh. Amin, Department of Pharmacognosy, Tehran University of Medical Sciences. A voucher specimen (No. 6509-T) was preserved at the herbarium of Faculty of Pharmacy, Tehran University of Medical Sciences. The aerial parts of plant were dried in the shade and powdered so that all the material could be passed through a mesh not larger than 0.5 mm.

2.2. Extraction procedures

Powdered plant material (600 g) was soaked in 1 l of methanol for 1 day, and the step was repeated twice, followed by filtration through filter paper. The filtrate was evaporated to dryness under reduced pressure and weighed (51.84 g, 8.64%).

2.3. Pharmacological studies

2.3.1. Carrageenan-induced rat paw edema

The anti-inflammatory activity of extract was determined by the carrageenan-induced edema test in the hind paws of rats. Male Albino Wistar rats (nine per group), 150–200 g, were fasted for 24 h before the experiment with free access to water. Fifty microlitres of a 1% suspension of carrageenan (Sigma Co., USA) in saline was prepared 1 h before each experiment and was injected into the plantar side of both hindpaws of the rats. The aerial parts extract and indomethacin were suspended in tween 80 plus 0.9% (w/v) saline solution. The final concentration of tween 80 did not exceeded 5% and did not cause any detectable effect per se. The extract at the doses of 4, 10, 20, 40, 80, 160 and 200 mg/kg and indomethacin at the doses of 2, 4, 6, 8 and 10 mg/kg were administered intraperitoneally (i.p.). Drugs or drugless vehicle were injected 1 h before the carrageenan treatment. Paw volume was measured immediately after carrageenan injection and at 1-, 2-, 3- and 4-h intervals after the administration of the edematogenic agent using a plethysmometer (model 7159, Ugo Basile, Varese, Italy). The degree of swelling induced was evaluated by the ratio a/b, where a and b are total volumes of both hind paws after and before carrageenan treatment, respectively. A ratio smaller than 1.5 after drug administration was considered as a significant inhibitory effect of the drugs (Chi and Jun, 1990).

2.3.2. Formalin test

Male Albino Wistar rats (nine per group), 150–200 g, were kept in plexiglas cages with free access to food and water. Testing took place in the middle of the light period of a 12-h light:12-h dark cycle. Each animal was tested once only. Plant extract (60, 70, 80 and 90 mg/kg) and morphine sulfate (5 and 10 mg/kg) were suspended in tween 80 plus 0.9% (w/v) saline solution and administered i.p. in a volume of 1.5–2 ml. Control group received only drugless vehicle (1.5–2 ml). The antinociceptive activity of the drugs was determined using the formalin test described by Dubuisson and Dennis (1977). One hour before testing, the animal was placed in a standard cage ($30 \times 12 \times 13$ cm), that served as an observation chamber. 50 μl of 2.5% formalin injected to the dorsal surface of the left hindpaw. The rat was observed for 60 min after the injection of formalin, and the amount of time spent licking the injected hindpaw was recorded. The first 5 min post formalin injection is known as the early phase and the period between 15 and 60 min as the late phase. The drugs were administered 15 min before injection of formalin.

2.3.3. Hot plate test

Male Swiss mice were placed on an aluminum hot plate kept at a temperature of 55 ± 0.5 °C for a maximum time of 30 s (Franzotti et al., 2000). Reaction time was recorded when the animals licked their fore- and hind paws and jumped; at before (0) and 15, 30, 45 and 60 min after intraperitoneal administration of 60, 70, 80 and 90 mg/kg of extract to different groups of eight animals each. Morphine 10 mg/kg was used as the reference drugs.

2.3.4. Rotarod test

The integrity of motor coordination was assessed with a rotarod apparatus, at a rotating speed of 16 r.p.m. A preliminary selection of mice was made on the day of experiment excluding those that did not remain on the rotarod bar for two consecutive periods of 45 s each (Pieretti et al., 1999). The number of falls from the rod was counted for 45 s (male Swiss mice 20–25g, nine per group). The performance time was measured before and 15, 30, 45 and 60 min after extract administration.

2.3.5. Acute toxicity

LD_{50} was assumed using 50% death within 72 h following i.p. administration of the extract at different doses (400, 504, 634, 798, 1004, 1264 and 1591 mg/kg). Male Swiss mice weighing 20–25 g (20 per group) were used in this experiment. The number of animals, which died during this interval, was expressed as a percentile, and the LD_{50} determined by probit test using a death percent versus doses' log (Carvalho et al., 1999).

2.3.6. Statistical analysis

ANOVA followed Student–Newman–Keuls test was used to determine significant differences between groups and $P < 0.05$ was considered significant.

Table 1
Anti-inflammatory effect of *G. grandiflorum* extract (GG) and Indomethacin on carrageenin-induced rat paw edema (3 h after carrageenan injection)

Treatment	Dose mg/kg (i.p.)	N	Degree of swelling $(a/b) \pm$ S.E.M.	Edema inhibition (%)
Control	–	9	1.99 ± 0.010	0.00
GG	4	9	1.72 ± 0.032	28.33 ± 3.232
	10	9	1.57 ± 0.030	42.66 ± 2.995
	20	9	1.42 ± 0.024	58.33 ± 2.427
	40	9	1.27 ± 0.021	73.11 ± 2.098
	80	9	1.23 ± 0.021	77.33 ± 2.075
	160	9	1.07 ± 0.021	92.44 ± 2.076
	200	9	1.01 ± 0.022	98.88 ± 2.176
Control	–	9	1.99 ± 0.010	0.00
Indomethacin	2	9	1.71 ± 0.023	28.88 ± 2.342
	4	9	1.38 ± 0.028	61.88 ± 2.831
	6	9	1.28 ± 0.023	71.88 ± 2.318
	8	9	1.21 ± 0.022	78.44 ± 2.193
	10	9	1.14 ± 0.021	86.00 ± 2.095

Student–Newman–Keuls analysis shows that all test groups are significantly from control group ($P < 0.001$).

3. Results and discussion

The carrageenan test is highly sensitive to non-steroidal anti-inflammation drugs, it has long accepted as a useful phlogistic tool for investigating new anti-inflammatory drugs (Just et al., 1998). The results obtained with extract and indomethacin in the carrageenan-induced edema test are shown in Table 1. The degree of swelling of the carrageenan-injected paws was maximal 3 h after injection and the mean increase in volume at that time was about 100% ($a/b \sim 2.0$) in control group. Statistical analysis shows that the edema inhibition of extract at the doses of 4, 10, 20, 40, 80, 160 and 200 mg/kg are significantly different from control group ($P < 0.001$). The degree of swelling (a/b) 3 h after carrageenan injection was ≤ 1.5 at the doses of 20–200 mg/kg.

The results showed that extract at the dose of 200 mg/kg had more edema inhibition than indomethacin at the doses of 10 mg/kg ($P < 0.01$) and 8 mg/kg ($P < 0.001$). The difference of edema inhibition (%) at the doses of 200 and 160 mg/kg of the extract was no significant ($P > 0.05$). Results indicated that similar activity against carrageenan-induced rat paw edema was observed with extract at the dose of 40 mg/kg and indomethacin at the dose of 6 mg/kg. In this study, extract exhibited a considerable anti-inflammatory activity ($ED_{50} = 13.59$ mg/kg) in comparison with drug reference.

Effects of extract on formalin test have been shown in Fig. 1. Statistical analysis shows that all test groups are significantly different from control group on the early and late phases ($P < 0.001$). The extract (60–90 mg/kg, i.p.) caused graded inhibition of both phases of formalin-induced pain. Antinociceptive activity of the extract at the doses of 80 and 90 mg/kg in the early phase was more than morphine at the dose of 5 mg/kg and less than morphine at the dose of 10 mg/kg ($P < 0.001$). Anti-nociceptive activity of the extract at the doses of 70, 80 and 90 mg/kg in the late phase was more than morphine at the dose of 5 mg/kg and less than morphine at the dose of 10 mg/kg ($P < 0.001$). The response of the extract at the doses of 80 and 90 mg/kg had not statistically significant difference to each other on the early and late phases pain response ($P > 0.05$).

Table 2
Effect of *G. grandiflorum* extract on the latency time of mice submitted to the hot plate test

Group	Dose (mg/kg)	N	Latency (s)				
			0 (min)	15 (min)	30 (min)	45 (min)	60 (min)
Control	–	8	6.40 ± 0.025	6.46 ± 0.011	6.01 ± 0.026	5.98 ± 0.025	5.55 ± 0.023
Morphine	10	8	6.42 ± 0.013	11.33 ± 0.028	13.1 ± 0.039	13.21 ± 0.026	13.30 ± 0.019
GG	60	8	6.39 ± 0.011	7.72 ± 0.017	7.94 ± 0.016	8.31 ± 0.015	6.56 ± 0.018
	70	8	6.40 ± 0.011	7.90 ± 0.016	8.12 ± 0.017	8.49 ± 0.013	6.69 ± 0.011
	80	8	6.42 ± 0.009	9.22 ± 0.018	9.38 ± 0.016	10.61 ± 0.019	6.89 ± 0.019
	90	8	6.42 ± 0.010	9.23 ± 0.016	9.39 ± 0.015	10.63 ± 0.017	6.91 ± 0.018

Each group represents the mean $\pm$ S.E.M. for eight animals. $P < 0.001$ compared with corresponding control values determined by Student–Newman–Keuls test.

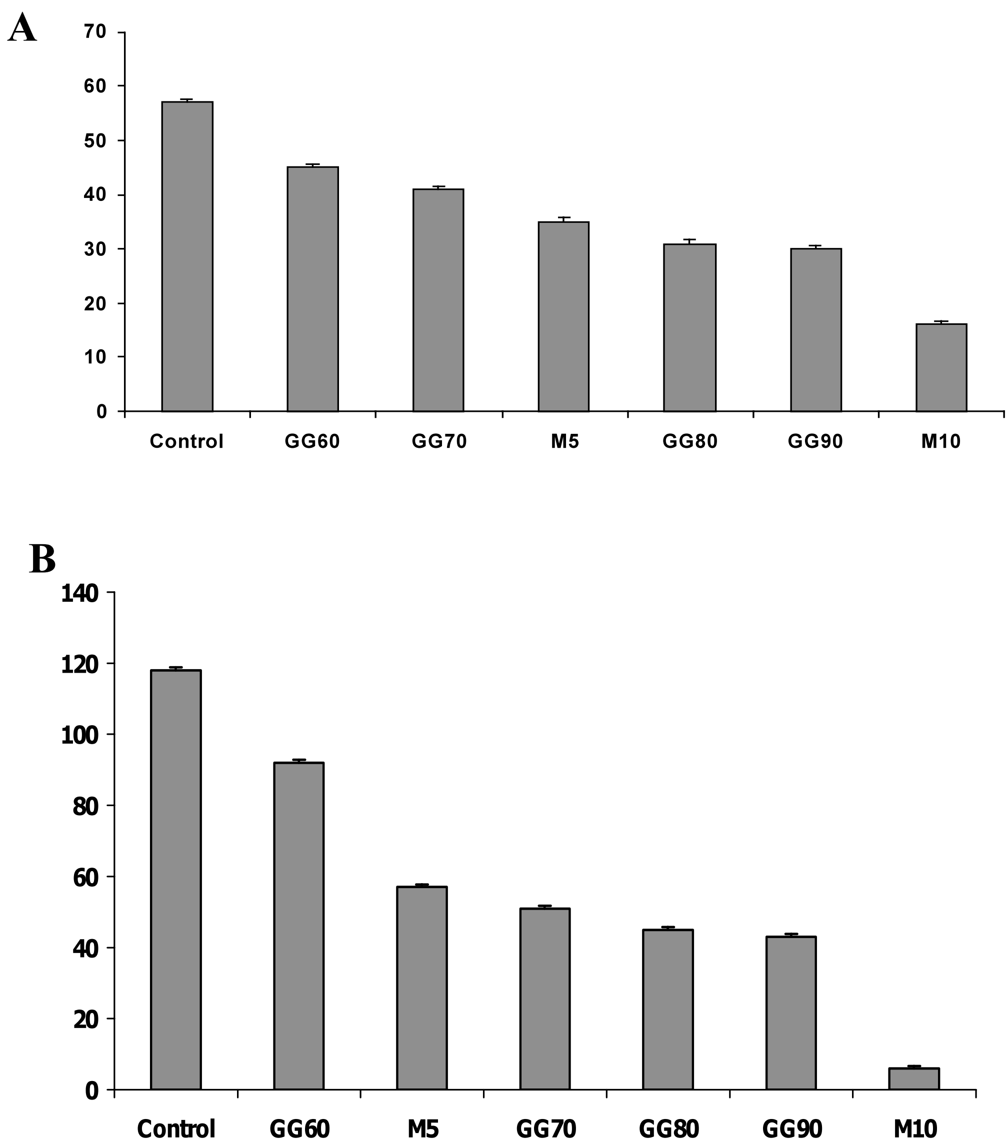

Fig. 1. Effect of *G. grandiflorum* extract (GG 60, 70, 80 and 90 mg/kg) and morphine (M 5 and 10 mg/kg) on the scores of the first phase (A) and the second phase (B) of the formalin test. The scores were calculated during a period of 300 s (0–5 min after formalin injection) for the first time and a period of 2700 s (15–60 min after formalin injection) for the second phase ($n = 9$ in each group). Values represent the mean ± S.E.M. of nine experiments. Student–Newman–Keuls analysis shows that all test groups are significantly different from control group ($P < 0.001$).

Table 3
Effects induced by *G. grandiflorum* extract (GG) administrated i.p. in the rotarod test

Treatment	Dose	Number of falls in 45 s (mean ± S.E.M.)			
		Minutes after extract treatment			
		15	30	45	60
Control	–	1.90 ± 0.010	1.10 ± 0.017	0.79 ± 0.025	0.90 ± 0.020
GG	60	1.88 ± 0.007	1.09 ± 0.017	0.83 ± 0.015	0.89 ± 0.022
	70	1.88 ± 0.012	1.08 ± 0.015	0.83 ± 0.017	0.88 ± 0.019
	80	1.87 ± 0.019	1.05 ± 0.012	0.78 ± 0.024	0.87 ± 0.025
	90	1.86 ± 0.018	1.05 ± 0.023	0.76 ± 0.022	0.87 ± 0.015

Each group represents the mean ± S.E.M. for nine animals.

The formalin test is a valid and reliable model of nociception and is sensitive for various classes of analgesic drugs. Formalin test produced a distinct biphasic response and different analgesics may act differently in the early and late phases of this test.

Therefore, the test can be used to clarify the possible mechanism of antinociceptive effect of a proposed analgesic (Tjolsen et al., 1992). Centrally acting drugs such as opioids inhibit both phases equally (Shibata et al., 1989) but Peripherally acting drugs such as aspirin, indomethacin and dexamethasone only inhibit the late phase. The late phase seems to be an inflammatory response with inflammatory pain that can be inhibited by anti-inflammatory drugs (Hunskaar and Hole, 1987; Rosland et al., 1990). The effect of extract on the first and second phases of formalin test suggests that its activity may be resulted from its central action when compared with morphine activity in this respect.

Based on the results of this study, we suggest that the antinociceptive effect of extract may be attributed to inhibition of prostaglandin release and other mediators involved in this test (Farsam et al., 2000). Hot plate test was also assayed to characterize the analgesic activity of extract. The results presented in Table 2 show that the i.p. administration of the extract at the doses of 60, 70, 80 and 90 mg/kg significantly raised the pain threshold at observation time of 45 min in comparison with control ($P < 0.001$). Morphine, used as a reference drug, also produced a significant antinociceptive effect during all the observation times when compared with control values ($P < 0.001$). The hot plate test is considered to be selective for opioid-like compounds in several animal species, but other centrally acting drugs, including sedatives and muscle relaxants, have also shown activity in this test (Hiruma-Lima et al., 2000). The hot plate test measures the response to a brief, noxious stimulus; the formalin test, on the other hand, measures the response to a long-lasting nociceptive stimulus, and thus may bear a closer resemblance to clinical pain (Rosland et al., 1990).

Anti-inflammatory and analgesic activity may also be related partly to the glaucine (isolated also in this work) and tetrahydropalmatine alkaloids reported for this plant.

Anti-inflammatory effects of the extract on the carrageenan test suggest that antinociceptive effect of the extract in the second phase of formalin test could have been partially due to its peripheral action. The extract at the doses of 20–200 mg/kg caused a significant inhibition during the 3rd h that is the phase of prostaglandin release (Franzotti et al., 2000). Thus, it seems that extract relieved pain through both central and peripheral mechanisms.

Further we have also tested the analgesic doses of extract for its effect on motor coordination by rotarod test. The results obtained are reported in Table 3. The *G. grandiflorum* extract (60–90 mg/kg) did not display any significant effect on the motor coordination of animals when tested in the rotarod test.

The 72 h acute LD_{50} value of this extract after i.p. administration in mice was 797.94 mg/kg.

Further studies are necessary to elucidate the mechanism behind its traditional effects. The results of pharmacological tests performed in the present study suggest that the extract is of low toxicity and presents analgesic and anti-inflammatory effects.

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